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ALTERNATING SIGN ROLLING TECHNOLOGY IN GROOVED ROLLS FOR NONFERROUS METAL PLATE BILLETS

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Results are given for a study of rolling plate metal by technology including rolling in grooved rolls and levelling on a smooth barrel. Results of computer modeling and analysis of the Lode–Nadai coefficient show that during rolling in grooved rolls two types of deformation-compression arise in a billet cross section, i.e., compression in inclined areas and shear in ridged zones. This picture of stress distribution is most favorable since compressive strain leads to "healing" of possible metal discontinuities, but shear deformation leads to intense working of the original metal structure. During rolling in the first pair of smooth rolls deformation compressive in nature predominates, and in the second pair of rolls it is deformation alternating in nature. In any billet vertical section, metal contact with rolls occurs on both surfaces, i.e., upper and lower, which leads to compression in ridge zones, but to tension in inclined sections. Analysis of the results of metallographic studies points to the benefits for rolling in grooved rolls in the plan of preparing a metal fine-grained isotropic structure. The use of grooved rolls leads to significant strengthening of L63 brass compared with using smooth rolls with an identical overall degree of deformation.

Keywords: *rolling, steel plate, grooved rolls, modeling, microstructure.*

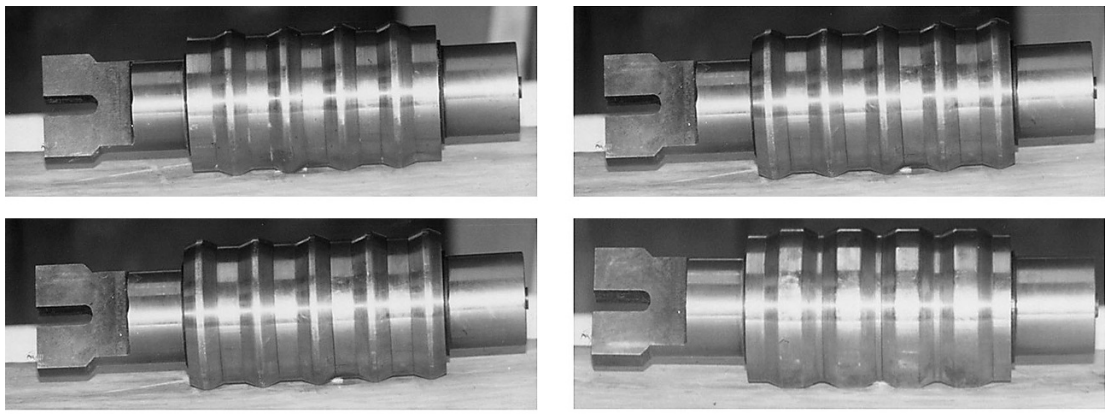
In the contemporary stage of the metallurgical industry development, attention of scientists and technologists in industrial enterprises is devoted to developing and studying new methods for preparing high quality competitive products. The use of classical plate rolling technology having a uniform nature of deformation to manufacture product with a prescribed set of properties involves additional heat treatment or correction of alloy composition, which leads to an increase in finished product cost. A promising direction for improving hot-rolled sheet properties, and subsequently in cold-rolled sheet, is development of production rolling processes making it possible to provide an increase in strength properties alongside good ductility by grain refinement for original metal in the course of implementing a deformation scheme, realizing rapid alternating-sign deformation throughout the whole volume of the deformed metal.

It is well known that one of the main ways of achieving alternating-sign deformation during various metal forming processes in an increase in the shear component of deformation (macroshear). Therefore, an important condition for improving the efficiency of rolled plate production is the use of local deformation providing creation within workpieces of significant accumulated alternating deformation, and also its inhomogeneity [1]. In order to realize these conditions (impossible with classical longitudinal rolling) in practice, it is necessary the change the shape of rolling rolls or a workpiece. One of the main methods of shape change is application of alternating ridges and recesses in the surface of the broad faces of rolls or workpieces [2, 3].

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a *b*

Fig. 1. Grooved rolls with unequal (*a*) and equal (*b*) ratio of projection width to recess width.

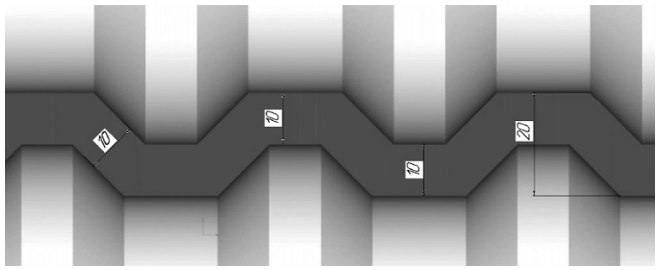


Fig. 2. Grooved roll layout.

This change in roll shape (or workpiece) makes it possible to create additional metal flow not only in a longitudinal, but also in a transverse direction to the rolling axis, which correspondingly leads to intensification of shear deformation throughout the whole volume of deformed metal and also to a change in metal property anisotropy. However, often in practice this method is connected with a marked change in the original workpiece dimensions, which is undesirable and even has an unfavorable effect since in a number of cases a significant change in cross section dimensions (normally a reduction) leads to a sharp reduction in the possible range of products prepared from a workpiece. In addition, a marked change in original workpiece dimensions leads to considerable energy expenditure which is also undesirable.

In view of this, it is important to develop new or improved technology with the aim of creating the most favorable conditions for preparing high quality metal plate without marked change in workpiece original dimensions.

Taking account of the advantages of rolling with additional shear deformation, the authors have developed an energy saving rolling technology for plate, making it possible to resolve this problem, and have designed two pairs of working rolls of new construction. This is rolls with a grooved surface in the form of circular channels forming trapezoidal ridges and recesses alternating over the whole barrel length [4]. The first pair is with an unequal ratio of ridge to recess width (Fig. 1*a*), and the second pair with an equal ratio of the width of ridge to width of recess (Fig. 1*b*) making it possible in practice to implement the prosed technology.

The advantages have already been given in [5] for rolling a plate workpiece in grooved rolls of the suggested construction (see Fig. 1) compared with smooth rolls, and it has been demonstrated that the optimum construction is grooved rolls with an unequal width of ridges to width of recesses (Fig. 1*a*). In this work, little attention was devoted to evaluating the forming scheme of deformation arising during workpiece rolling by the proposed technology and a study of the study of the effect of rolled surface relief on the metal microstructure of plate workpieces.

The aim for the authors of this work is computer modeling of the developed technology using the Simufact Forming software for studying workpiece shape change during rolling and evaluation of the deformation scheme during workpiece rolling by the proposed technology, and also a study of the effect of this scheme on the evolution of metal microstructure.

In order to study workpiece shape change during rolling and to evaluate the deformation scheme workpiece rolling by the proposed technology, the authors developed a geometric model of grooved rolls with an uneven ratio of the width of projections to the width of recesses (Fig. 2) corresponding entirely to a pair of rolls of a laboratory duo 100 mill. The diameter of the grooved roll over the collar D is 100 mm, barrel length L_b = 150 mm, and the recess depth equals the ridge height $h_{re} = h_{ri}$ = 10 mm.

The computer model of the suggested technology was composed using a finite element method with the Simufact Forming software. The models consists of a stand with grooved rolls and two successively installed stands with smooth rolls intended for levelling a workpiece after rolling in grooved rolls. The entire rolling process in the model is separated into three stages: rolling in grooved rolls; primary levelling of a workpiece in rolls with a smooth barrel; secondary workpiece levelling. The overall form of the model is given in Fig. 3.

The original workpiece used was sheet of rectangular shape with a size $h \times b \times l$ = 10 × 140 × 200 mm. The workpiece starting was brass grade L63.

The conditions and assumptions adopted during modeling in the Simufact Forming program were:

1) workpiece in the original condition (before deformation) isotropic and initial stresses and deformation absent from it;

2) rolling wad conducted at 20°C;

3) a tool is absolutely rigid;

4) starting workpiece material taken as ductile;

5) a Siebel friction model adopted, i.e., contact stress exceeds yield strength;

6) the friction coefficient between tool and workpiece is selected on the basis of recommendations for the Simufact Forming program and taken as 0.7; and

7) rolling rate is 1.25 rad/sec.

In the course of studying modeling results, workpiece shape change was analyzed during rolling plate by the prescribed technology (Fig. 4).

After rolling in the first stand, sheet has alternating ridges and recesses of trapezoidal shape, identical to the roll shape. Workpiece thickness is 20 mm (see Fig. 4*b*). In the second pass, there is levelling of a workpiece over the cross section as a result of metal flow in the direction of least resistance, i.e., in a direction due to the curved profile (see Fig. 4*c*). There is reduction in ridge height, and in this case workpiece thickness remains unchanged. Workpiece thickness was 15 mm.

In the next pass, there is final leveling of ridges with insignificant workpiece broadening. It may be concluded from the modeling results obtained that successive installation of two pairs of smooth rolls makes it possible to level out a workpiece after passing through grooved rolls (see Fig. 4*d*). Workpiece thickness is 10 mm.

The results obtained point to the fact that the shape of workpiece cross section after two passes in smooth rolls returns to rectangular without a marked change in original workpiece dimensions.

In studying the stress-strained state, a possibility develops not only of evaluating the stresses and strains arising in a workpiece, but also to determine the overall type of deformation at a specific point. As a result of performing the analysis, it is possible to determine what type of deformation predominates at metal deformation sites, i.e., tension, compression, or shear.

As a criterion for evaluating the deformation scheme, the Lode–Nadai coefficient was used [6], calculated by the equation

$$\mu = 2(\sigma_2 - \sigma_3)/(\sigma_1 - \sigma_3) - 1,$$

where σ_1, σ_2, and σ_3 are the main stresses (MPa) in all three rolling stages.

Values of the principal stresses are taken from results of computer modeling of the process. For the calculation, values of maximum, average, and minimum principal stresses were taken at 28 points over the whole length of the cross section (see Fig. 4). Values of Lode–Nadai coefficient are in the range from 1 to –1. Values of the coefficient tending towards one point to the compressive nature of deformation, towards zero they point to a shear nature, and to –1 they point to a stretching nature [6].

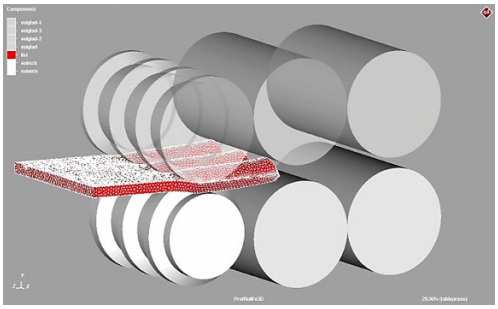
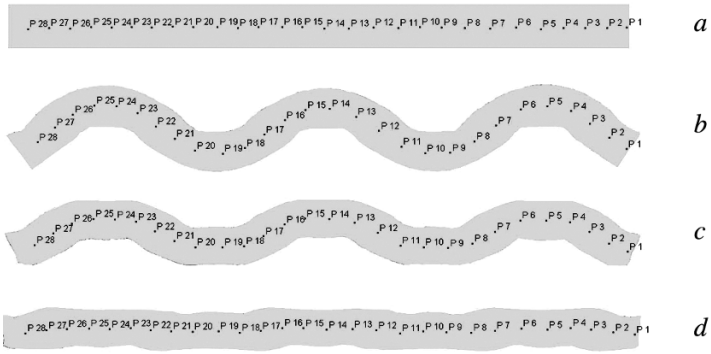


Fig. 3. Overall view of model.



a

b

c

d

Fig. 4. Billet shape change with respect to passes: *a*) original billet; *b*) after rolling in grooved rolls; *c*) after second pass (smooth rolls); *d*) after third pass (smooth rolls).

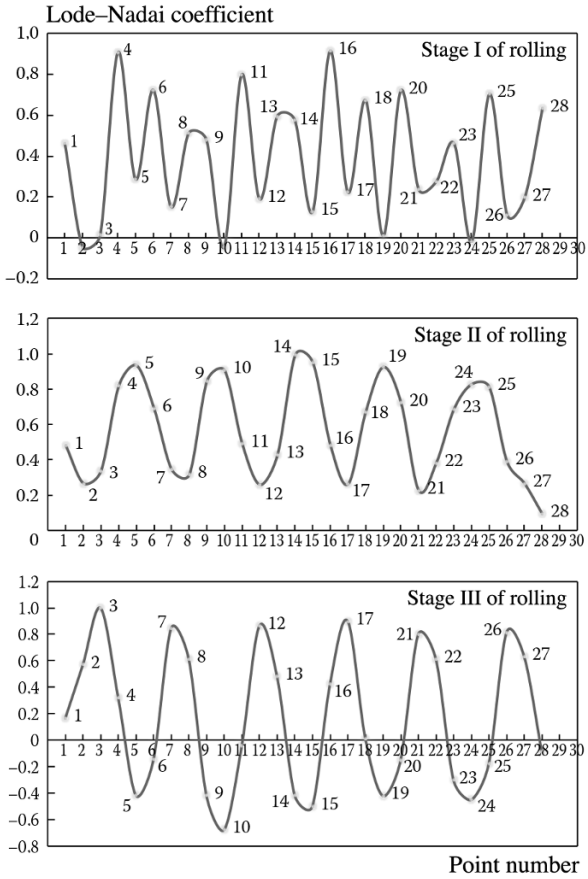


Fig. 5. Results of Lode-Nadai coefficient calculation.

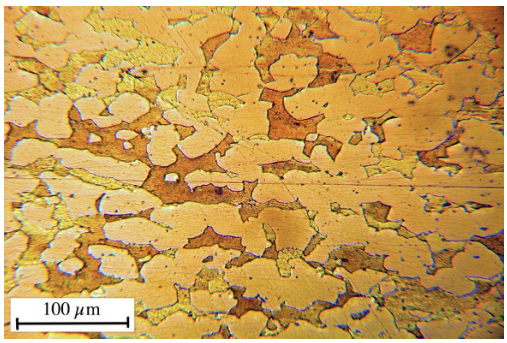
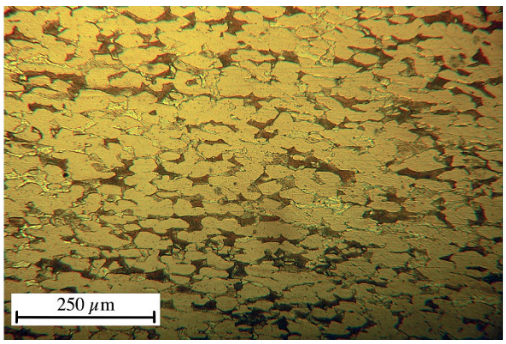
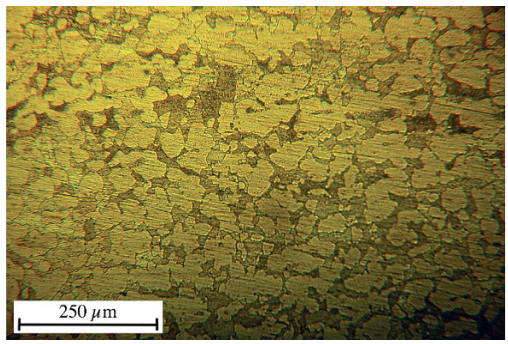


Fig. 6. Brass L63 structure after annealing at 700°C.

a





b

Fig. 7. Brass L63 structure after rolling by existing (*a*) and suggested (*b*) technology.

Results obtained in the course of analyzing the Lode–Nadai coefficient (Fig. 5) showed that in the first rolling stage in the grooved rolls within a workpiece cross section two types of deformation arise, i.e., compression and shear, which is indicated by the value of this coefficient, i.e., positive tending towards 0. Shear deformation is realized in ridge zones, whereas compression arises in sections adjacent to them. This picture of stress distribution is most favorable since compressive deformation leads to "healing" of possible metal discontinuities, and shear deformation leads to rapid working of the original metal structure.

In the second rolling stage (smooth rolling), shear deformation predominates; therefore, the values of the coefficient are positive are in the range 0.2–1. This phenomenon may be explained as follows. After the grooved rolls, the height of ridges (relief) of a workpiece increases with unchanged thickness. On entry into the smooth rolls, there is workpiece levelling out over the thickness, i.e., workpiece straightening. In this case, a typical distribution of values of the coefficient is observed, i.e., in inclined sections of a workpiece it is at a minimum (0.2–0.3); in zones of ridges, metal contact arises with rolls and therefore compressive stresses are significantly greater and the value of the coefficient is at a maximum (0.9–1).

Deformation in the third rolling stage is alternating in nature. Here, in contrast to the two preceding stages, in any vertical section of a workpiece metal contact with rolls occurs on both surfaces, i.e., upper and lower, which leads not only to compression in ridged zones but also to extension in inclined sections. This is explains insignificant workpiece broadening. In this case, a very typical distribution is also observed for the coefficient, i.e., within inclined workpiece sections it is negative, –(0.4–0.3), which points to the presence of tensile deformation; in ridged zone, metal contact occurs with both rolls, and therefore here the value of the coefficient is at a maximum (0.9–1).

As already noted above, preparation of finished rolled product with good mechanical properties is one of the main tasks of any metallurgical process, including manufacture of metal product from copper and its alloys. Currently, in industrial enterprises of Kazakhstan and Russia producing metal product of copper and its alloys, which concerns the Balkhash Plant for Nonferrous Metal Treatment, an increase in mechanical properties of these alloys is mainly achieved by adjustment of

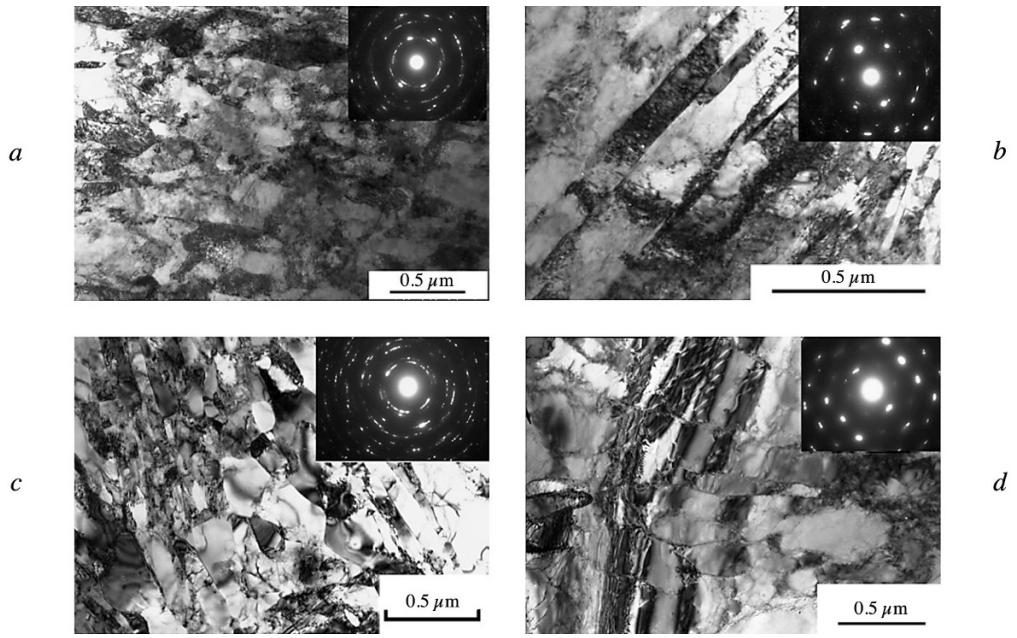


Fig. 8. Microstructure of brass billets after rolling, SEM-study: *a*) in smooth rolls, transverse direction; *b*) in smooth rolls, longitudinal direction; *c*) in grooved rolls with unequal ratio of ridge width to recess width, transverse direction; *d*) in grooved rolls with unequal ratio of ridge width to recess width, longitudinal direction.

alloy composition using alloying additions or by finished rolled product heat treatment. In order to improve the quality of the product, the authors have suggested introducing into the Balkhash Plant instead of existing strip and coil rolling technology for copper and its alloys the rolling technology developed, i.e., with the use of alternating-sign deformation. The strip and coil rolling technology existing in the plant for copper and its alloys includes rolling in a universal rolling mill. Feature of the stand is the fact that it operates both in a two-roll regime (duo) and also in a four-roll regime (quarto).

In order to determine the possibility of using grooved rolls in the production of rolled product at the Balkhash OTsM Plant, the most widespread alloy in industry, i.e., brass L63, was chosen. Test rolling by the technology proposed was performed in a duo 100 laboratory mill under Karaganda State Industrial University conditions. Workpieces were prepared for study with a size of 10 × 100 × 300 mm. Before the start of experiments, specimens were annealed in a muffle furnace at 700°C for 1 h. Workpiece rolling was performed by the proposed technology in grooved rolls with an uneven ratio of the width of projection to width of recess, and by existing technology in smooth rolls. Rolling in grooved rolls with an uneven ratio of projection to recess width was accomplished in five passes with a reduction in each pass $\Delta h = 0.6$ mm. After which stagewise workpiece deformation was performed in smooth rolls to a final workpiece thickness equal to 5 mm.

A second batch of workpieces was rolled in a duo 100 laboratory mill by the technology existing at the Balkhash Plant with a reduction regime 10 mm → 8.5 mm → 7.2 mm → 6.3 mm → 5.8 mm → 5.4 mm → 5.0 mm.

Then, specimens were studied in the original (annealed) condition and after rolling by two different regimes, and their hardness and microstructure were determined. Alloy microstructure is shown in Fig. 6 after annealing at 700°C. It is represented by broken equiaxed grains of α-solid solution between which there are quite fine layers of β-phase, i.e., as a result of heating a significant part of the CuZn intermetallic dissolved in solid solution based on copper. In this case, an increase in possible in alloy ductility and its capacity for cold deformation.

In spite of high-temperature annealing, within the alloy structure there are signs of certain directionality in the position of both phases, which may point to a graded action on it of high degrees of deformation or nonequilibrium crystallization with strengthen heat removal by mold material. The absence of twins, typical for previously deformed and annealed alloy, points to the fact that the original condition is cast.

Fine inclusions of impurities appear as a result of annealing, which are located both within the α-phase grains themselves, which does not affect ductility, and also over their boundaries. There is a stable tendency of solid solution grain intergrowth based on copper and formation of low-angle boundaries typical for a polygonised structure. Signs of recrystallization and formation of α-phase annealing twins was not established.

The microstructure of brass L63 after rolling by the proposed and existing technologies is shown in Fig. 7. In the course of studying the microstructure of the surface of brass workpieces, it was revealed that the original grain size is refined during deformation.

Workpieces rolled in smooth rolls have clearly defined grain elongation in the rolling direction, and consequently property anisotropy within the volume of deformed metal. Analysis of the surface of workpieces rolled in grooved rolls with an uneven ratio of width of ridges and recesses shows that grain extension was not detected either in the longitudinal or cross sections. Use of this tool makes it possible to prepare fine-grained isotropic structure within the volume of deformed metal.

Studies in GEOL JEM2100 transmission electron microscope showed that rolling in both smooth and grooved rolls leads to preferential formation of dislocation lines and slip bands in the longitudinal direction. In this case, the maximum difference in amount of slip lines with comparison of longitudinal and transverse sections is observed after rolling brass workpieces in smooth rolls (Fig. 8). Apart from this, use of smooth rolls leads to nonuniform sliding of dislocations since they only have the most favorable orientated slip systems. Visually, this is expressed as the fact that within individual grains dislocation slip lines are greater than in others (se Fig. 8*a*, *b*). After rolling in grooved rolls, this is not observed, the distribution of slip lines in individual grains and one or another direction is uniform (Fig. 8*c*, *d*), i.e., sliding dislocations operate to an equal extent in all grains, and consequently there is plastic deformation.

In addition, grooved rolls give yet another advantage compared with smooth rolls: during microscope study of a specimen surface with in both longitudinal and transverse directions dislocation slip lines are seen arranged at an angle to each other. Consequently, slip proceeds through different planes and directions. In order to achieve this effect with the use of smooth rolls, it is necessary to increase the degree of deformation, which leads to an increase in energy consumption. Apart from this, during workpiece deformation in smooth rolls metal breakdown may occur before the stage of multiple slip, since with an identical dislocation density sliding in this case proceeds through a smaller number of planes than with use of grooved rolls. Consequently, in each individual plane more individual dislocations move, which by collecting around a barrier leads to microcrack formation. During workpiece deformation in grooved rolls, more slip planes operate, in each plane there are fewer slip lines, and additional energy is unnecessary for creating a critical dislocation density, which leads to the formation of discontinuities.

Values of Vickers HV (H) for brass L63 are given below:

After annealing	8725
After rolling smooth rolls	17254
After rolling in grooved rolls	19316

The use of grooved rolls causes an increase in rolled brass strength compared with that in smooth rolls.

Conclusions. The results of computer modeling of rolling thick sheet metal by the suggested technology, including rolling in grooved rolls and levelling workpieces on a smooth barrel, and values of Lode–Nadai coefficient obtained in the course of analysis, indicate that rolling in grooved rolls in a transverse section of a workpiece two types of deformation arise, i.e., compression in inclined areas and shear in ridge zones. This picture of stress distribution is most favorable since compressive deformation leads to "healing" of possible metal discontinuities, and shear deformation leads to intense working of the original metal structure.

Shear deformation predominates in the second rolling stage in smooth rolls. Deformation in the third rolling stage is alternating in nature. Here, in any vertical workpiece section metal contact arises in rolls at both surfaces, i.e., upper and lower, which leads not only to compression in zones of ridges, but also to extension in inclined sections. This explains the insignificant workpiece broadening.

Combined analysis of the results of metallographic studies points to preferred application of grooved rolls compared with smooth rolls from the point of view of obtaining an isotropic fine-grained rolled metal structure, and consequently

improved plate quality. The use of grooved rolls also facilitates significant strengthening of brass L63 during rolling compared with use of smooth rolls with an identical overall degree of deformation.

On the basis of this, it is possible to recommend introduction of grooved rolls with an uneven ratio of width of projection with width of recess in metallurgical enterprises manufacturing rolled metal sheet of nonferrous metals and alloys.

REFERENCES

1. E. G. Pashinskaya and A. V. Zavdoveev, "Use of rolling with shear and traditional drawing for forming the structure and properties of low-carbon steels," *Fiz. Tekhn. Vysok. Temp.*, **22**, No. 3, 113–123 (2012).
2. A. I. Traino and V. P. Polukhin, "Rolled sheet manufacture using local deformation," *Proizv. Prokata*, No. 1, 25–31 (2011).
3. A. V. Brinza, V. P. Polukhin, V. V. Brinza, et al., USSR Inventor's Cert. 1088818, IPC[3] B21B1/22, "Rolling method," subm. 05.20.1983, publ. 04.30.1984.
4. A. B. Naizabekov and S. N. Lezhnev, Kazakhstan Republic Patent 14791, IPC B21B27/02, "Rolls for rolling thick sheets," subm. 03.06.2003, publ. 11.15.2003, *Byull.*, No. 11.
5. A. Naizabekov, S. Lezhnev, T. Koinov, et al., "Research and development of technology for rolling of high-quality plates of non-ferrous metals and alloys in relief rolls," *J. Chem. Technol. Metallurgy*, **51**, No. 4, 363–370 (2016).
6. V. M. Segal, V. I. Reznikov, V. I. Kopylov, et al., *Metal Plastic Structure Formation Processes*, Nauka i Tekhnika, Kiev (1994).